

Chapter 5: Further Applications of Newton's Laws

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PHY201
Lecture 5



Part I

Friction

Outline of Part I: Friction

Introduction to Friction

Solving Friction Problems

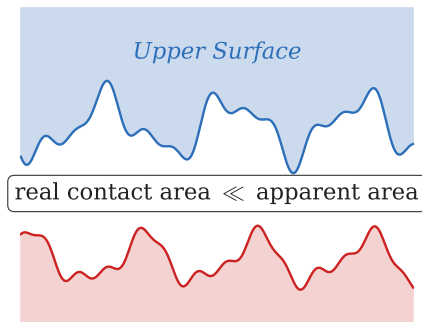
Friction: A Closer Look

Friction ($\vec{f}$) — SI Unit: N

In Chapter 4, friction was introduced as a contact force opposing relative motion.

In Chapter 5, we quantify friction using the **coefficient of friction** μ and the **normal force** N .

- ▶ Friction arises from electromagnetic interactions between surface atoms.
- ▶ Proportional to how hard the surfaces are pressed together.
- ▶ **Independent of contact area** (within limits).



Two Types of Friction

Kinetic Friction (f_k)

Acts when surfaces are **sliding** against each other:

$$f_k = \mu_k N$$

- ▶ μ_k : coefficient of kinetic friction
- ▶ N : normal force magnitude
- ▶ f_k is **constant** during sliding

Static Friction (f_s)

Acts when surfaces are **not** sliding — prevents motion from starting:

$$f_s \leq \mu_s N$$

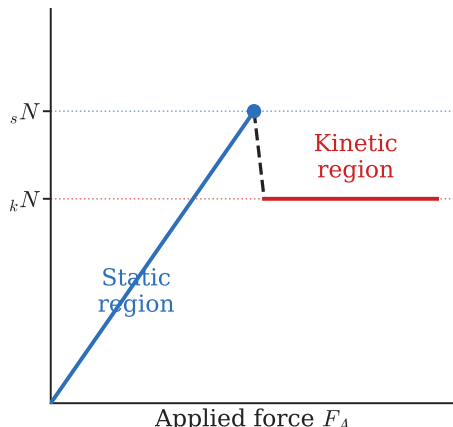
- ▶ μ_s : coefficient of static friction
- ▶ f_s **adjusts** to match the applied force
- ▶ Maximum: $f_{s,\max} = \mu_s N$

Always: $\mu_s \geq \mu_k$ — it is harder to **start** sliding than to **keep** sliding.

Static Friction: Adjustable Up to a Maximum

Static friction is not a fixed value. It **adjusts** to exactly oppose the applied force — up to its maximum $f_{s,\max} = \mu_s N$.

Applied force F_A	Static friction f_s
2 N	2 N — at rest
5 N	5 N — at rest
9 N	9 N — barely
10 N	object starts to slide!
(Here $\mu_s N = 10 \text{ N}$)	



Coefficients of Friction

The coefficients of friction depend only on the **materials** in contact — not on area, speed, or mass.

System	μ_s	μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.7	0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Steel on steel (dry)	0.6	0.3
Bone (synovial fluid)	0.016	0.015
Teflon on Teflon	0.04	0.04
Shoes on ice	0.1	0.03

$\mu_s \geq \mu_k$ in every case. Anti-lock brakes keep tires rolling (static friction) rather

Check Your Understanding

A book rests on a table. You push it with 3 N of force and it does not move.

What is the friction force on the book?

- (a) 0 N
- (b) 3 N
- (c) $\mu_s N$
- (d) $\mu_k N$

Come to lecture to see the answer.

Check Your Understanding

Why is it easier to keep a heavy box sliding than to start it sliding?

Answer in terms of μ_s , μ_k , and N .

Come to lecture to see the answer.

Outline of Part I: Friction

Introduction to Friction

Solving Friction Problems

How to Solve a Friction Problem

Strategy (extends the Ch. 4 four-step approach)

1. Draw the FBD. Identify all forces, including $\vec{f}$.
2. Determine whether the object is **moving or stationary**:
 - ▶ Moving $\Rightarrow$ use $f_k = \mu_k N$
 - ▶ Stationary $\Rightarrow$ use $f_s \leq \mu_s N$; find f_s from $\sum \vec{F} = 0$
3. Apply $\sum \vec{F} = m\vec{a}$ in each direction.
4. Solve for the unknown.

Never assume $N = mg$. Recalculate N from the FBD every time — it changes on inclines and when forces have vertical components.

Example: Billy Pushes a Box (Part a)

Minimum Force to Start Sliding — (cp2e_ch5)

Billy pushes a 2.00 kg box horizontally across the floor.

$$\mu_s = 0.100, \quad \mu_k = 0.0500$$

- (a) What is the minimum horizontal force F_A needed to start the box sliding?

$$m = 2.00 \text{ kg}, \quad \mu_s = 0.100$$

$$\sum F_y = 0 : \quad N = mg = (2.00)(9.80) = 19.6 \text{ N}$$

$$f_{s,\max} = \mu_s N = (0.100)(19.6 \text{ N})$$

$$F_{A,\min} = f_{s,\max} = 1.96 \text{ N}$$

To start sliding, the applied force must overcome maximum static friction:

$$F_{A,\min} = \mu_s N$$



Example: Billy Pushes a Box (Part b)

Acceleration Once Sliding — (cp2e_ch5)

Billy continues to push with the same force $F_A = 1.96 \text{ N}$.

The box is now **sliding**.

(b) Find the acceleration of the box.

$$\mu_k = 0.0500, \quad N = 19.6 \text{ N}$$

$$f_k = \mu_k N = (0.0500)(19.6 \text{ N}) = 0.980 \text{ N}$$

$$\sum F_x = ma : \quad F_A - f_k = ma$$

$$1.96 - 0.980 = (2.00) a$$

$$a = 0.490 \frac{\text{m}}{\text{s}^2}$$

$f_k < f_{s,\max}$ even though F_A unchanged.

The net force is now nonzero; the box accelerates.

Switch from μ_s to μ_k as soon as the object begins moving.

Check Your Understanding

A 10.0 kg block rests on a horizontal surface.

$$\mu_s = 0.400$$

What is the minimum horizontal force needed to start the block sliding?

Come to lecture to see the answer.

Check Your Understanding

A 5.00 kg block slides down a frictionless incline at $\theta = 30.0^\circ$.
Now add kinetic friction: $\mu_k = 0.200$.
Find the new acceleration.

Come to lecture to see the answer.

Part II

Drag Forces and Terminal Velocity

Outline of Part II: Drag Forces and Terminal Velocity

Drag Force

Drag Force: Review and Extension

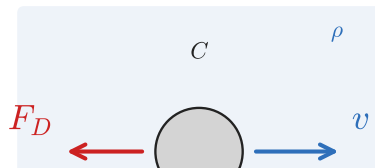
Drag Force (F_D) — SI Unit: N

From Chapter 4, the drag force is the retarding force a fluid exerts on a moving object:

$$F_D = \frac{1}{2} C \rho A v^2$$

where C is the drag coefficient, ρ is the fluid density ($\frac{\text{kg}}{\text{m}^3}$),
 A is the cross-sectional area (m^2), and v is the speed relative to the fluid.

- ▶ $F_D \propto v^2$: doubling speed **quadruples** drag
- ▶ $F_D \propto A$: larger cross-section $\Rightarrow$ more drag
- ▶ $F_D \propto \rho$: denser fluid $\Rightarrow$ more drag



Stokes' Law: Drag at Low Speeds

Stokes' Law (F_s) — SI Unit: N

For **small, slow-moving** spheres in a viscous fluid, drag is proportional to v :

$$F_s = 6\pi\eta rv$$

where η is the dynamic viscosity (Pa·s), r is the sphere's radius (m), and v is the speed ($\frac{\text{m}}{\text{s}}$).

Fluid	η (mPa·s)
Water (20°C)	1.00
Blood	2.7–4.0
Olive oil	81
Motor oil	100

Use Stokes' law when flow is **laminar** (smooth): small object, low speed.

Use $F_D = \frac{1}{2}C\rho Av^2$ for large, fast objects (skydivers, baseballs) where flow is **turbulent**.

Terminal Velocity: Extended Analysis

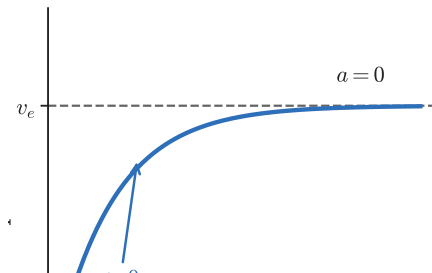
Terminal Velocity (v_e)

Recall from Chapter 4: at terminal velocity, $F_D = w$, so acceleration is zero:

$$\frac{1}{2} C \rho A v_e^2 = mg \quad \Rightarrow \quad v_e = \sqrt{\frac{2mg}{C \rho A}}$$

The Ch. 4 squirrel reached $v_e = 9.24 \frac{\text{m}}{\text{s}}$. What about a skydiver?

- ▶ Large A (parachute) $\Rightarrow$ low v_e
- ▶ Large m $\Rightarrow$ high v_e
- ▶ Denser fluid $\Rightarrow$ lower v_e



Example: Skydiver Terminal Velocity

Skydiver in Free Fall — (cp2e_ch5)

A skydiver (with gear) has total mass $m = 80.0$ kg.
 Spread-eagle position: $C = 1.00$, $A = 0.700$ m², $\rho_{\text{air}} = 1.225$ $\frac{\text{kg}}{\text{m}^3}$.
 Find the terminal velocity.

$$m = 80.0 \text{ kg}, \quad A = 0.700 \text{ m}^2$$

$$v_e = \sqrt{\frac{2mg}{C\rho A}} = \sqrt{\frac{2(80.0)(9.80)}{(1.00)(1.225)(0.700)}}$$

$$v_e = \sqrt{\frac{1568}{0.8575}} = \sqrt{1828}$$

$$v_e = 42.8 \frac{\text{m}}{\text{s}} \approx 154 \frac{\text{km}}{\text{h}}$$

$$\approx 96 \frac{\text{mi}}{\text{h}}$$

Opening a parachute increases A by $\sim 100\times$, dropping v_e to $\sim 5 \frac{\text{m}}{\text{s}}$.

Parachute: $A \sim 70 \text{ m}^2$
 $v_e \propto 1/\sqrt{A}$, so $100\times$ larger A gives $10\times$ lower v_e .

Check Your Understanding

The skydiver above opens a parachute, increasing effective area from 0.700 m^2 to 70.0 m^2 .

By what factor does terminal velocity change?

Come to lecture to see the answer.

Check Your Understanding

A tiny spherical dust particle falls slowly through air at very low speed.

Which drag law applies: $F_D = \frac{1}{2} C \rho A v^2$ or $F_s = 6\pi\eta r v$?

Explain your reasoning.

Come to lecture to see the answer.

Part III

Elasticity, Stress, and Strain

Outline of Part III: Elasticity, Stress, and Strain

Hooke's Law

Elastic Moduli

Hooke's Law

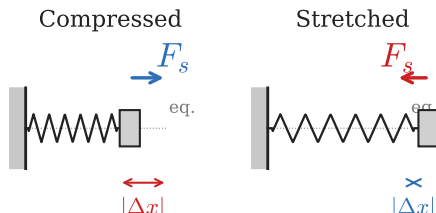
Hooke's Law

When a spring is displaced by Δx from equilibrium, it exerts a **restoring force**:

$$F_s = -k \Delta x$$

where k is the **spring constant** ($\frac{\text{N}}{\text{m}}$) and Δx is the displacement. The negative sign means the force **opposes** the displacement.

- ▶ Large $k \Rightarrow$ stiff spring (large force for small Δx)
- ▶ Small $k \Rightarrow$ soft spring
- ▶ Valid only within the **elastic limit** — beyond it, the material deforms permanently



Stress and Strain

Stress and Strain

Stress Force per unit area applied to a material:

$$\text{stress} = \frac{F}{A} \quad (\text{Pa})$$

Strain Fractional change in the relevant dimension (dimensionless):

$$\text{strain} = \frac{\Delta L}{L_0} \quad \text{or} \quad \frac{\Delta x}{L_0} \quad \text{or} \quad \frac{\Delta V}{V_0}$$

The **elastic modulus** is the ratio of stress to strain — a measure of material stiffness:

$$\text{elastic modulus} = \frac{\text{stress}}{\text{strain}}$$

Outline of Part III: Elasticity, Stress, and Strain

Hooke's Law

Elastic Moduli

Young's Modulus: Tension and Compression

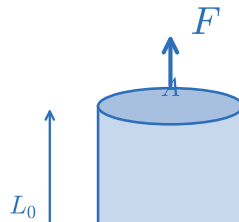
Young's Modulus (Y) — SI Unit: Pa

Describes resistance to **stretching or compressing** along an axis:

$$\Delta L = \frac{1}{Y} \frac{F}{A} L_0$$

where ΔL is the change in length, F/A is the applied stress, and L_0 is the original length.

- ▶ Large $Y \Rightarrow$ stiff material (small ΔL)
- ▶ Small $Y \Rightarrow$ flexible material (large ΔL)
- ▶ $\Delta L > 0$: stretching; $\Delta L < 0$: compression



Shear Modulus and Bulk Modulus

Shear Modulus (S)

Resistance to **shear** (parallel forces):

$$\Delta x = \frac{1}{S} \frac{F}{A} L_0$$

Δx : lateral displacement

Forces act **parallel** to surface.

Tearing paper, twisting a rod, cutting with scissors.

Bulk Modulus (B)

Resistance to **uniform compression** (pressure from all sides):

$$\Delta V = \frac{1}{B} \frac{F}{A} V_0$$

ΔV : change in volume

Pressure applied **uniformly**.

Deep-sea vessel compressed by water pressure.

Elastic Moduli of Common Materials

Elastic moduli are properties of the **material**, independent of object shape.

Material	Y (GPa)	S (GPa)	B (GPa)
Steel	210	80	170
Aluminum	70	25	75
Copper	120	48	140
Bone (tensile)	16	80	—
Marble	60	20	70
Glass	70	30	37
Rubber	0.0001	0.0002	2
Water	—	—	2.2

Steel is $\sim 10^6 \times$ stiffer than rubber in tension, yet both obey $\Delta L = \frac{1}{Y} \frac{F}{A} L_0$.

Example: Steel Cable Under Load

Elongation of a Steel Cable — (cp2e_ch5)

A steel cable ($Y = 2.10 \times 10^{11} \text{ Pa}$) has length $L_0 = 2.00 \text{ m}$ and cross-sectional area $A = 1.00 \times 10^{-4} \text{ m}^2$.

A load $F = 2000 \text{ N}$ is suspended from it.

Find the elongation ΔL .

$$Y = 2.10 \times 10^{11} \text{ Pa}, \quad L_0 = 2.00 \text{ m}, \quad F = 2000 \text{ N}$$

$$\begin{aligned} \Delta L &= \frac{1}{Y} \frac{F}{A} L_0 \\ &= \frac{(2000)(2.00 \text{ m})}{(2.10 \times 10^{11} \text{ Pa})(1.00 \times 10^{-4} \text{ m}^2)} \end{aligned}$$

Stress:

Strain:

$$\frac{\Delta L}{L_0} = \frac{F}{YA}$$

$$\Delta L = 1.90 \times 10^{-4} \text{ m} = 0.190 \text{ mm}$$

Check Your Understanding

A bone has $Y = 1.60 \times 10^{10} \text{ Pa}$, length $L_0 = 0.400 \text{ m}$, and cross-sectional area $A = 4.00 \times 10^{-4} \text{ m}^2$.

A compressive force $F = 1200 \text{ N}$ is applied.

Find $|\Delta L|$.

Come to lecture to see the answer.

Check Your Understanding

Which type of elastic deformation describes each scenario?

- (a) Squeezing a rubber ball uniformly from all sides.
- (b) Pulling on both ends of a rubber band.
- (c) Sliding the top face of a gelatin cube sideways.

Come to lecture to see the answer.

Next Lecture

Chapter 6: Uniform Circular Motion and Gravitation

- ▶ Uniform circular motion: centripetal acceleration $a_c = v^2/r$
 - ▶ Centripetal force: $F_c = mv^2/r$
 - ▶ Newton's Law of Universal Gravitation
 - ▶ Satellites and orbital velocity
-
- ▶ Newton's Second Law still applies — centripetal force is just a net force directed inward.
 - ▶ The FBD strategy from Ch. 4 works for circular motion problems too.